

WORKED EXAMPLE

Read the exact schedule and its index

Lecture4 PDF page57 supplies these three formulas.

$$\eta_k = \frac{\eta_0}{k+1}, \quad \eta_k = \frac{\eta_0}{(k+1)^2}, \quad \eta_k = \eta_0 e^{-\beta k}$$

Eta-zero is the starting learning rate. k is the update index, beginning at zero. The rate at index k moves the current parameter to the next one.

The slide calls the first schedule linear decay and the second quadratic decay. Their denominators are linear and quadratic in k+1: both are reciprocal schedules.

In particular, linear here does not mean subtract a fixed amount from the rate. Beta is a positive exponential decay constant, separate from any momentum coefficient.

$$\eta_0 = 0.12, \quad \beta = \ln 2$$

$$w_{k+1} = w_k - \eta_k g(w_k)$$

WORKED EXAMPLE

Reciprocal linear: substitute k before dividing

Every denominator uses $k+1$, including the first update.

$$k = 0 : \quad \eta_0 = 0.12/(0 + 1) = 0.12$$

$$k = 1 : \quad \eta_1 = 0.12/(1 + 1) = 0.06$$

$$k = 2 : \quad \eta_2 = 0.12/(2 + 1) = 0.04$$

$$k = 3 : \quad \eta_3 = 0.12/(3 + 1) = 0.03$$

The drops are 0.06, 0.02 and 0.01, so the rate does not decrease by a constant amount. Starting at $k=1$ would incorrectly skip the original rate.

WORKED EXAMPLE

Reciprocal quadratic: square the whole denominator

Parentheses determine which quantity is squared.

$$k = 0: \quad \eta_0 = 0.12/(0 + 1)^2 = 0.12$$

$$k = 1: \quad \eta_1 = 0.12/(1 + 1)^2 = 0.12/4 = 0.03$$

$$k = 2: \quad \eta_2 = 0.12/(2 + 1)^2 = 0.12/9 \approx 0.01333333$$

$$k = 3: \quad \eta_3 = 0.12/(3 + 1)^2 = 0.12/16 = 0.0075$$

At $k=2$ the denominator is 9, not k -squared plus 1 = 5. It decays faster than the reciprocal-linear schedule with the same initial rate.

WORKED EXAMPLE

Exponential: use the logarithm identity

Beta=ln2 means multiplying by one half at each update.

$$e^{-k \ln 2} = 2^{-k} \implies \eta_k = 0.12(2^{-k})$$

$$k = 0: \quad 0.12(2^0) = 0.12$$

$$k = 1: \quad 0.12(2^{-1}) = 0.12/2 = 0.06$$

$$k = 2: \quad 0.12(2^{-2}) = 0.12/4 = 0.03$$

$$k = 3: \quad 0.12(2^{-3}) = 0.12/8 = 0.015$$

The natural logarithm ln2 is the number whose exponential is 2. A negative exponent makes the multiplier smaller as k increases.

WORKED EXAMPLE

Compare the first four scheduled rates

All three agree at initialization, then decay differently.

k	reciprocal linear	reciprocal quadratic	exponential
0	0.12	0.12	0.12
1	0.06	0.03	0.06
2	0.04	0.01333333	0.03
3	0.03	0.0075	0.015

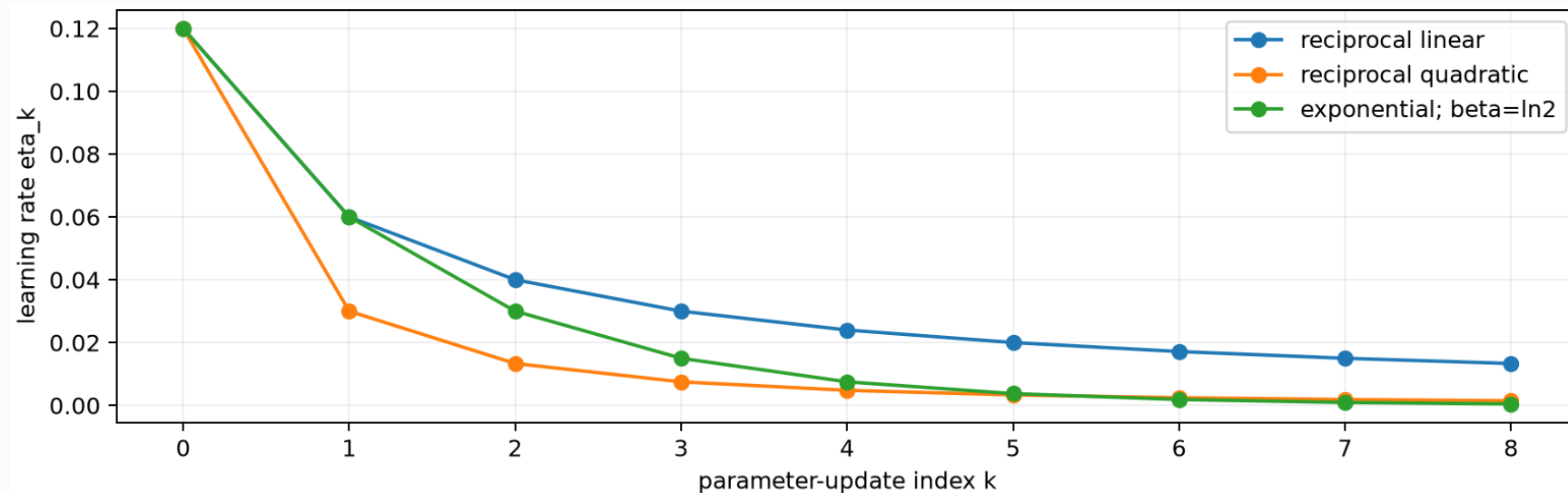
These are alternative schedules. Do not apply all three successively to one rate unless a different rule explicitly says to combine them.

A rate table is not a parameter table: actual parameter changes also depend on the current gradient.

WORKED EXAMPLE

See the rates over more update indices

The plotted points include $k=0$ rather than starting at $k=1$.



Exponential decay has a constant multiplicative ratio. The reciprocal schedules have ratios that vary with the update index. A lower rate does not automatically imply better training.

WORKED EXAMPLE

Apply the rate with the matching current gradient

Use only reciprocal-linear decay in this worked parameter trace.

$$E(w) = \frac{1}{2}w^2, \quad g(w) = w, \quad w_0 = 1$$

$$k = 0: \quad \eta_0 = 0.12, \quad w_1 = 1 - 0.12(1) = 0.88$$

$$k = 1: \quad \eta_1 = 0.06, \quad w_2 = 0.88 - 0.06(0.88) = 0.8272$$

$$k = 2: \quad \eta_2 = 0.04, \quad w_3 = 0.8272 - 0.04(0.8272) = 0.794112$$

Notice both the rate and the gradient change. At update $k=2$, use the gradient at w_2 , then obtain w_3 . The first rate is not reused for every step.

WORKED EXAMPLE

A plateau rule responds to measured progress

The lecture says reduce the rate when training or held-out performance stagnates.

For this study example, measure validation loss after each epoch. An epoch means a pass through the training examples; its count here begins at 1, unlike the update index k .

Chosen rule: improve only if the new validation loss is strictly below the best seen so far. After two consecutive epochs without improvement, multiply the current rate by 0.1.

Reset the no-improvement counter after a new best or a rate drop. The reduced rate takes effect in the next epoch. Retain the learned parameters.

$$0.12(0.1) = 0.012, \quad 0.012(0.1) = 0.0012$$

This precise patience rule is an illustrative choice. Page57 gives the stagnation-triggered idea and multiplicative reduction, not these exact counter settings.

WORKED EXAMPLE

Trace the plateau decision at every epoch

Best is the lowest validation loss observed; smaller is better.

epoch	val loss	best	counter/action	next rate
1	0.500	0.500	new best; 0	0.12
2	0.450	0.450	new best; 0	0.12
3	0.460	0.450	1	0.12
4	0.470	0.450	2: drop; reset to 0	0.012
5	0.440	0.440	new best; 0	0.012
6	0.445	0.440	1	0.012
7	0.446	0.440	2: drop; reset to 0	0.0012

Epoch 4 used rate 0.12; its decision sets rate 0.012 for epoch 5. The decision after epoch 7 sets rate 0.0012 for epoch 8. No earlier update is changed retroactively.

INDEPENDENT PRACTICE

Your turn: new initial rate and decay constant

Calculate the scheduled rate before updating a supplied parameter state.

$$\eta_0 = 0.2, \quad k = 2, \quad \beta = \ln 4$$

Calculate reciprocal-linear, reciprocal-quadratic and exponential rates at $k=2$. Show the substituted denominator or exponent.

$$E(w) = \frac{1}{2}w^2, \quad \text{supplied } w_2 = 2$$

Using only the reciprocal-quadratic rate, calculate w_3 . This w_2 is an independent supplied practice state, not the result of the earlier main-example updates.

Explain the difference between an update-index schedule and the epoch-based plateau rule.

WORKED ANSWER

Practice: rates and the next parameter

Use $k+1=3$ before squaring the denominator.

$$\eta_{2, linear} = 0.2/(2 + 1) = 0.2/3 \approx 0.066666667$$

$$\eta_{2, quadratic} = 0.2/(2 + 1)^2 = 0.2/9 \approx 0.022222222$$

$$\eta_{2, exp} = 0.2e^{-2\ln 4} = 0.2(4^{-2}) = 0.2/16 = 0.0125$$

$$g(w_2) = 2, \quad w_3 = 2 - \frac{0.2}{9}(2) = 2 - \frac{0.4}{9} \approx 1.955555556$$

The formula schedules change with k regardless of measured validation progress. The plateau rule changes only after its stated no-improvement trigger; it keeps the current learned parameters.